
ENERGY SAVING, NEW, AND RENEWABLE ENERGY SOURCES

Thermochemical Recovery of Heat Contained in Flue Gases by Means of Bioethanol Conversion

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Abstract—In the present paper consideration is being given to the use of bioethanol in the schemes of thermochemical recovery of heat contained in exit flue gases. Schematic diagrams illustrate the realization of thermochemical heat recovery by implementing ethanol steam conversion and conversion of ethanol by means of products of its complete combustion. The feasibility of attaining a high degree of recovery of heat contained in flue gases at the moderate temperature (up to 450°C) of combustion components is demonstrated in the example of the energy balance of the system for thermochemical heat recovery. The simplified thermodynamic analysis of the process of ethanol steam conversion was carried out in order to determine possible ranges of variation of process variables (temperature, pressure, composition) of a reaction mixture providing the efficient heat utilization. It was found that at the temperature above 600 K the degree of ethanol conversion is near unity. The equilibrium composition of products of reaction of ethanol steam conversion has been identified for different temperatures at which the process occurs at the ratio $H_2O/EtOH = 1$ and at the pressure of 0.1 MPa. The obtained results of calculation agree well with the experimental data.

Keywords: biofuel, renewable energy sources, degree of conversion, thermochemical heat recovery, heat transformation, energy efficiency, ethanol

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In recent years a considerable amount of attention has been devoted to the study of problems related to production and use of alternative organic fuel—biofuel. One of the types of such organic fuel is ethanol (bioethanol). The body of scientific publications, in which problems related to the use of bioethanol, are discussed increases steadily, and in the last decade, according to the SCOPUS bibliographic database containing abstracts and citations for academic journal articles, it has increased by a factor of about 3.

Such considerable interest in bioethanol is in no way accidental, because even today bioethanol is used widely in internal combustion engines, air-breathing jet engines, “small” power industry, etc. [1]. It is logical to imagine that within a few years bioethanol may find an important place in the “large” power industry as well, for example, at thermal power plants, heat-generating process installations, etc.

The problems of increasing the effectiveness of the use of bioethanol as fuel remain to be topical. One of the promising directions in enhancing the available heat factor is thermochemical heat recovery, the principle of which is that heat contained in flue gases, is used for preliminary endothermal treatment of the original fuel. As a result of this treatment, there is a larger amount of chemically bonded energy in the form of an increased heating value. The latest investigations carried out by the author, as well as by other specialists, show that the thermochemical heat recovery

holds much promise, because even today it has found application in high-temperature heat-process installations [2, 3], gas turbine units [4], internal combustion engines [5], and so on.

This paper deals with the problems of combined implementation of two promising directions: the use of bioethanol in the large power industry and thermochemical heat recovery.

Methods of thermochemical heat recovery by means of methane steam conversion [2, 4] and methane conversion by products of its complete combustion [3, 6] have gained the widest usage. Despite the compelling advantage of these methods—attainment of a high level of heat recovery—they also have disadvantages: a limited range of use as to the flue-gas temperature (above 750°C) and the high probability of coke formation at the high-temperature conversion process. The preliminary estimates show that ethanol steam conversion and ethanol conversion by products of complete combustion in systems for thermochemical heat recovery will be possible at flue-gas temperatures in the range from 400 to 450°C, and, as a result, the probability of deposition of soot on the catalyst surfaces will decrease considerably.

Figure 1 shows the schematic diagrams of the use of thermochemical recovery of heat contained in flue gases by means of ethanol conversion.

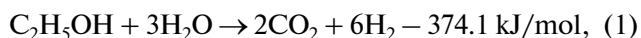
The process of transformation of chemical energy (heat of combustion) of ethanol for schemes shown in

Fig. 1 can be conditionally divided into two stages. The first stage is an increase in chemically bonded fuel energy in the form of increased heat of combustion (the heating value) due to preliminary endothermal treatment. The second stage is firing of the converted gas (CG) (hydrogen-containing gas), which has a higher heating value as compared to that of original substances.

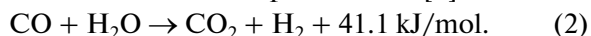
It must be emphasized that the first stage is of particular interest as the method of cooling (thermal protection) of thermally stressed constructional elements. Thermal protection is provided not only by means of heat removal by convection and radiation (at high enough temperatures of the surface being cooled), but also by means of heat absorption at phase and chemical transformations, and in the case of the latter, unlike ablation transformations that tend to destroy heat-protection materials, are not destructive processes, but serve as a method for producing new fuel [7].

Figure 1a shows the system for thermochemical heat recovery by means of ethanol steam conversion. Flue gases are directed for surface heating of the reaction chamber of a thermochemical reactor (TCR) in which reaction of ethanol steam conversion occurs. The reaction mixture consists of water and ethanol vapors, and before being fed into the reactor chamber the mixture is heated in the reaction mixture heater (RMH) by residual heat of flue gases. One of the disadvantages of such a scheme is the necessity of steam generation for ethanol conversion. It is possible to eliminate this disadvantage by using products of full combustion of ethanol for its conversion (see Fig. 1b). Flue gases downstream of the power plant are divided into two flows. One flow is directed for surface heating of the thermochemical reactor, while the other is mixed with ethanol in the reaction chamber of the reactor, and, as a result, ethanol conversion by products of its full combustion takes place. One of the essential disadvantages of such a scheme is the low pressure of the reaction mixture, with the result that the volume of the reaction chamber of the thermochemical reactor is larger for one scheme (b) than for the other scheme (a). Converted gas (hydrogen-containing gas) downstream of the thermochemical reactor is used as the fuel in a power plant.

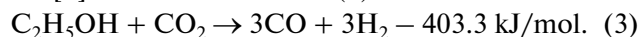
In the thermochemical reactor (scheme a) the following reaction of ethanol steam conversion occurs [8]:



which involves the water vapor reaction [8]:



For scheme b in the thermochemical reactor the chemical reaction of carbon dioxide ethanol conversion [9] is added to reaction (1):



Chemical processes in the thermochemical reactor for both schemes involve many side reactions, the mechanism of which depends on technological condi-

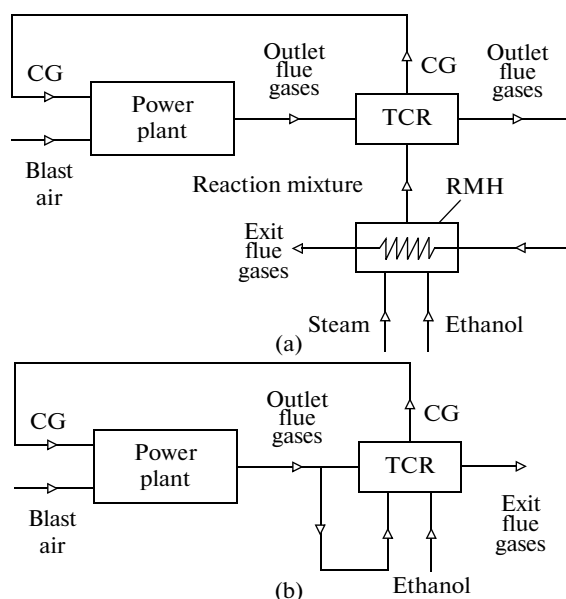


Fig. 1. Schematic diagram of the use of thermochemical recovery of heat contained in flue gases by ethanol steam conversion (a) and ethanol conversion by-products of complete combustion (b).

tions of the process occurrence (temperature, pressure, composition of a primary mixture), as well as on a type of the catalyst. Upon further thermodynamic analysis of the system for thermochemical heat recovery in the temperature interval from 300 to 800 K it was taken that reactions (1) and (3) are principal ones and that side reactions do not significantly affect the material and energy balance [8, 9].

Considering the prospects for the use of the thermochemical recovery of heat contained in flue gases by means of ethanol conversion, it is necessary, first of all, to note the feasibility of attaining a high level of recovery of heat contained in flue gases in the thermochemical reactor, and this is illustrated well by the example, which is considered hereafter. Besides, for a thermochemical reactor the specific quantity of metal per unit of heat recovered is considerably lower than that for conventional recuperative air heaters.

Conversion (see Fig. 1a) of 1 m³ of ethanol (C₂H₅OH) by the stoichiometric amount of water vapor (H₂O) yields 6 m³ of hydrogen (H₂) and 2 m³ of carbon dioxide (CO₂) according to reaction (1). In this case, an increase in chemically bonded energy is $\Delta Q_{\text{chem}} = 16.7 \text{ MJ/m}^3 \text{ EtOH}$. This fact indicates that the heating value of the original mixture C₂H₅OH + 3H₂O, equal to 22.31 MJ/m³ of EtOH, increased by 16.7 MJ/m³ of EtOH. Thus, the coefficient of transformation of sensible heat contained in combustion products into chemical energy of converted gas is equal to the ratio between the lower heating value of reaction products and the lower heating value of original substances, i.e., is equal to 1.75. At the final temperature of converted gas equal to 450°C its enthalpy

will be about 3.35 MJ per 1 m³ of original ethanol. In this case the total heat recovery in the thermochemical reactor will be 20.05 MJ/m³ of EtOH. If we reduce this value to 1 kg of original ethanol C₂H₅OH, we obtain about 9.8 MJ/kg of EtOH, which corresponds to the amount of heat contained in 9.52 m³ of air needed for combustion of 1 m³ of methane (with the excess air factor $\alpha = 1.0$), with the temperature of about 750°C. Thus, the degree of heat recovery that can be achieved in the system for thermochemical heat recovery by means of steam conversion of 1 kg of ethanol corresponds to the degree of heat recovery that can be achieved in conventional schemes by means of air heating to 750°C.

On the basis of this calculation it can be concluded that, owing to the use of a thermochemical reactor as a heat recovery system, it is possible to attain high recovery of heat contained in flue gases at a moderate temperature of combustion components (about 450°C.) Besides, since the gas mixture with hydrogen being its main component is used as fuel, hazardous emissions to the atmosphere [5] are reduced considerably, and a moderate temperature of the occurrence of the conversion process makes the formation of non-equilibrium carbon on catalytic surfaces less probable.

In the example given above it is taken conditionally that the degree of ethanol conversion in the thermochemical reactor is equal to 1, i.e., the reaction products do not contain original ethanol. In actual fact, the degree of ethanol conversion depends on process variables: temperature, pressure, and composition of the original reaction mixture. To determine this dependence, in the present work, the thermodynamic analysis of the chemical-technological process occurring in the thermochemical reactor has been carried out. The degree of ethanol conversion x_{EtOH} is found from the formula

$$x_{\text{EtOH}} = \frac{\text{EtOH}^{\text{in}} - \text{EtOH}^{\text{out}}}{\text{EtOH}^{\text{in}}} \times 100\%, \quad (4)$$

where EtOHⁱⁿ, EtOH^{out} are the amounts of ethanol at the reactor inlet and outlet.

The degree of ethanol conversion determines the enthalpy of flue gases that is transformed into chemical energy of converted gas. Determination of the value of x_{EtOH} is a vital issue when identifying prospects for implementing thermochemical heat recovery by means of ethanol conversion. The stoichiometric flow rate of reaction components (1) in the left-hand side of the equation $\Delta Q_{\text{chem}} = 374.1 \text{ kJ/mol}$, and in this case ΔQ_{chem} depends on the degree of ethanol conversion x_{EtOH} in the following way:

$$\Delta Q_{\text{chem}} = x_{\text{EtOH}} Q_{\text{end}(1)}, \quad (5)$$

where $Q_{\text{end}(1)}$ is the thermal effect of reaction (1) at the stoichiometric flow rate of the reacting component, kJ/mol.

It follows from the Le Chatelier–Brown principle that the high degree of ethanol regeneration x_{EtOH} in

endothermal reaction (1) can be attained by means of increasing the steam/ethanol ratio ($\beta = \text{H}_2\text{O}/\text{EtOH}$) in the feedstock flow and an increase in the temperature of the process occurring in the reactor. It is apparent that the temperature rise in the end part of the thermochemical reactor is of special importance for the extent to which reaction (1) has proceeded. If it is assumed that in a long enough section of the reactor adjacent to the gas exit the constant temperature is maintained (by means of organization of purposeful heat supply), it should be expected that in the reaction mixture the equilibrium ratio between concentrations of all reagents will be established. In terms of mathematical simulation such assumption means posing of an additional boundary condition at “semi-infinity,” i.e., at the infinite distance from the entrance of the flow into the system. The use of the approximation of a semi-infinite reactor makes it possible, on the basis of elementary equations of material balance alone, to carry out very important calculations of the degree of ethanol conversion that would be maximum (equilibrium) for each temperature level.

It is well known that at thermodynamic calculations of equilibrium compositions there is no sense in considering intermediate stages of a reaction, which are in a great number at ethanol steam conversion. Besides, it is believed that all molecules of elementary substances do not dissociate into atoms because of relatively low temperatures at which the process occurs, that all gases are ideal ones (in the sense that chemical potentials of the substances are determined by their partial pressures), and that in the reaction products no free carbon is contained and no complex hydrocarbons are formed there.

In order for the value of x_{EtOH} , to be determined, it is necessary and sufficient to solve (with respect to x_{EtOH}) the equation that describes the equilibrium of reaction (1):

$$K_p = \frac{p_{\text{CO}_2}^2 p_{\text{H}_2}^6}{p_{\text{EtOH}}^3 p_{\text{H}_2\text{O}}^3} = \frac{v_{\text{CO}_2}^2 v_{\text{H}_2}^6}{v_{\text{EtOH}}^3 v_{\text{H}_2\text{O}}^3} p^5, \quad (6)$$

where p_i is the partial pressure of the i th component, Pa; p is the total pressure in the chemico-technological system, Pa; $v_i = \frac{\dot{q}_i}{\sum \dot{q}_i}$ is a mole fraction of the i th component; and $\dot{q}_i$ is the amount of the i th component, mol.

The equilibrium constants are calculated by means of the expression for finding Gibbs energy for each temperature level T using data borrowed from [10]:

$$-G^0 = RT \ln K_p, \quad (7)$$

where R is the universal gas constant, J/(mol K); and T is the temperature at which the process occurs, K.

Gibbs energy for each of the components is found from the Gibbs–Helmholtz equation:

$$G = h - Ts, \quad (8)$$

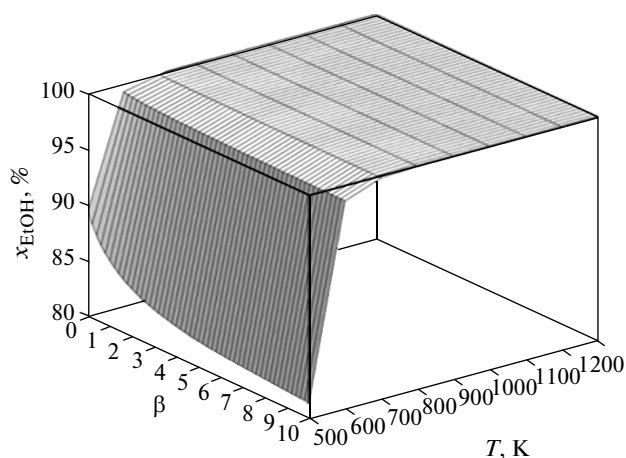


Fig. 2. The dependence of the degree of ethanol conversion on the temperature and the value of β at the inlet of the thermochemical reactor.

where

$$h = h^{298} + \int_{T=298}^T c_p dT; \quad s = s^{298} + \int_{T=298}^T \frac{c_p}{T} dT; \quad (9)$$

c_p is the specific heat at constant pressure, J/(mol K); s is the entropy, J/(mol K); and h is the enthalpy, J/mol.

The thermodynamic data needed for calculations from formulas (8) and (9) were borrowed from [10] as well.

In the present work calculations were carried out for the constant pressure in the system $p = 0.1$ MPa, and the ratio β varied from 0 to 10. Theoretically this ratio may be chosen arbitrarily, but at ratios much higher than stoichiometric values, the reaction products (converted gas) will contain a large amount of water vapor—a noncombustible component—and this fact will play a negative role when using converted gas as fuel. The range of temperatures at which the process occurs and which is analyzed here is from 300 to 1200 K.

Figure 2 illustrates the degree of ethanol conversion depending on the value of β at the inlet of the thermochemical reactor and on the temperature. On the basis of this dependence it may be concluded that at $T > 600$ K the degree of ethanol conversion is practically the maximum obtainable one and may be as high as 99% at various values of β . However, at low temperatures ($T < 500$ K) the degree of conversion decreases and depends on both the temperature and the ratio β . On the basis of this dependence it can be asserted that at $T < 500$ K the maximum possible coefficient of transformation of heat contained in flue gases into chemical energy will be attained. Figure 3 shows the dependence of the level of ethanol conversion on the temperature for different values of β ,

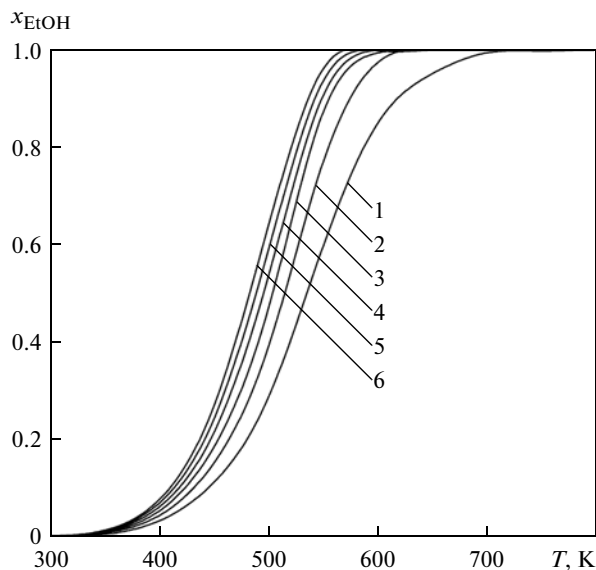


Fig. 3. The dependence of the degree of ethanol conversion on the temperature. Numbers of the curves correspond to the values of β .

which makes the three-dimensional image depicted in Fig. 2 clearer.

Figure 4 illustrates the dependence of the molar flow rate of hydrogen H_2 , carbon dioxide CO_2 , and ethanol $EtOH$ on the temperature at $\beta = 1$ and the pressure $p = 0.1$ MPa. The number of moles of ethanol and water is the same, according to the initial conditions; therefore, the graph of changing the molar flow rate of water is not shown in Fig. 4. Carbon dioxide is contained in conversion products due to the occurrence of reactions (1) and (2). It is well known that at temperatures below 750 K equilibrium of reaction (2) is shifted to the right, and, as a result, this reaction has a rather essential effect on the material balance of the process; therefore, with the rise in temperature, the amount of CO_2 will increase. It can also be seen from Fig. 4 that the number of moles of hydrogen at the degree of ethanol conversion near to unity is hardly higher than 5, whereas the number of moles of original ethanol is 1.3. This fact indicates that, apart from the chemical substances shown in Fig. 4, the conversion products contain some amount of by-products: ethylene (C_2H_4), acetaldehyde (C_2H_4O), etc. As a result, the number of moles of hydrogen for reaction (1) differs from the stoichiometric one. With the rise of temperature of the conversion process the yield of hydrogen per 1 mole of original ethanol increases. In addition, it seems worthwhile to bear in mind that, as noted above, the type of catalyst has an essential effect on the ultimate composition of the products of ethanol conversion by water vapor, and this fact is not considered in the thermodynamic analysis.

The presence of reaction by-products in the conversion products would alter the process balance, and,

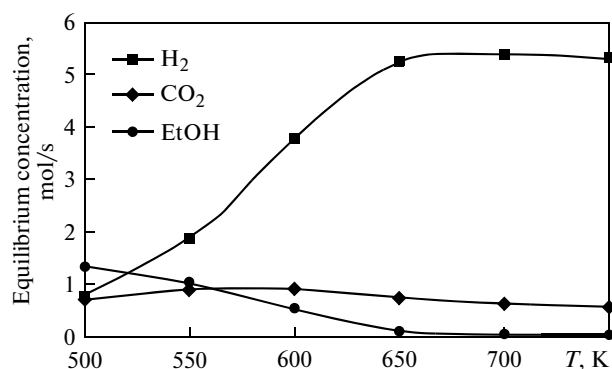


Fig. 4. Equilibrium concentrations of hydrogen H₂, carbon dioxide CO₂ and ethanol EtOH depending on the temperature at $\beta = 1$ and $p = 0.1$ MPa.

as can be seen from Fig. 4, the volume content of by-products may run up to 20% of the total volume of converted gas. These substances affect the energy balance only at a certain amount of sensible heat of converted gas, which in the example given above, was calculated under the condition that conversion products contain only hydrogen and carbon monoxide. By-products of reaction do not affect appreciably the total energy balance of the system for thermochemical heat recovery, because the amount of sensible heat contained in converted gas is 3.35 MJ per 1 m³ of ethanol, with the total amount of recovered heat equal to 20.05 MJ/m³ of EtOH. It has been known that the endothermal effect of common conversion reaction (1) depends on the degree of conversion found from (5); therefore, in the present study special attention was focused on the determination of just the equilibrium (maximum) degree of ethanol conversion.

In addition, as stated before, the selected type of catalyst affects strongly the composition of conversion products. Figure 5 shows the dependence of the volume concentration of different components of converted gas on the temperature [11], obtained by Hernandez with his coauthors at the experimental study of the process of ethanol steam conversion on nickel-tungsten catalyst Ni-W/Al₂O₃). It can be seen from Fig. 5 that the total volume of by-products at the conversion temperature of about 600 K is really about 20% of the total volume of converted gas. However, in the temperature range under study the main by-product is carbon dioxide, which exhibits chemical inertness in burning, while the content of ethylene (C₂H₄), acetaldehyde (C₂H₄O), and methane (CH₄) is extremely low.

It can be seen from the dependences obtained that at ethanol conversion by water vapor already at temperatures of about 600 K the degree of conversion close to the maximum value can be attained, whereas at methane steam conversion it is necessary to supply heat with the temperature of about 1000 K in order to achieve a degree of conversion equal to about unity [4,

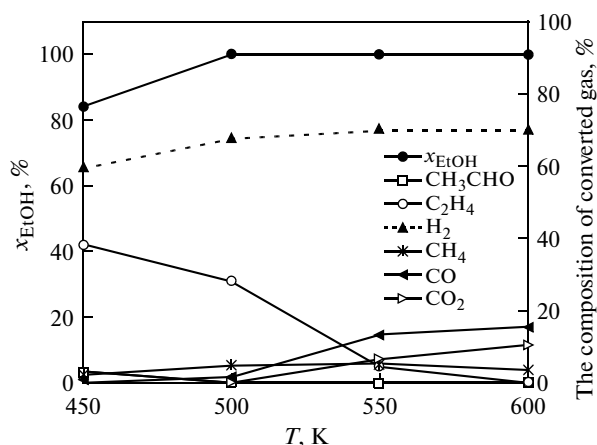


Fig. 5. The effect of the temperature at which the process occurs in the presence of the catalyst 30% Ni-W/Al₂O₃ on the composition of converted gas at $\beta = 2$ [11].

6]. Thus, the use of ethanol steam conversion for thermochemical heat recovery becomes feasible only in the cases when the temperature of flue gases is about 700 K, and this, in turn, leads to widening the scope of the implementation of this promising method of increasing the energy efficiency of fuel-fired installations. However, the thermodynamic analysis cannot lend full insight into the process of thermochemical heat recovery by means of ethanol conversion, but merely guides the directions in which scientific investigations should be developed. The further study of this process, in the author's opinion, holds much promise, because the principles of thermochemical heat recovery can be applied in virtually each power-generating facility that uses hydrocarbons, including bioethanol and other biofuels, as a fuel.

CONCLUSIONS

(1) At the flue gas temperature of about 600 K the degree of ethanol steam conversion equal to about unity can be attained; therefore one can assert that it is advisable to implement this method of heat recovery in power-generating facilities in which the flue gas temperature is about 600 K, and this can expand essentially the scope of the use of thermochemical heat recovery at industrial thermal power plants.

(2) The preliminary endothermal reprocessing of ethanol by means of steam conversion increases considerably the heating value of ethanol: the heating value of reaction products is by a factor of 1.75 higher than that of original ethanol. The preliminary endothermal reprocessing of ethanol makes it possible to use this type of biofuel, among other things, for the needs of high-temperature power generation processes.

(3) It is evident that the present-day level of the use of biofuel in our country gives no way of expecting that this method of the use of bioethanol in schemes of

thermochemical heat recovery will be implemented commercially in the coming 7–10 years. This is caused, first and foremost, by rather severe climatic conditions in Russia, which impede the production of low-cost ethanol despite the vast reserves of conventional organic hydrocarbon raw materials. However, the trends in the development of the present-day energy industry of the world point to an increase in the proportion of alternative fuels in the balance of the thermal power industry, and this fact makes it possible to speak in favor of the future prospects of this method of thermochemical heat recovery.

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